

Peer Review under Present Conditions

What already exists in the IPI exchange, in Dot Theory governance and in the FFGFT record-keeping, and what of it can be used for a review framework

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Contents

Peer Review under Present Conditions	2
Legend	2
1 Starting point	3
What the classical procedure assumes	3
What has changed	3
The question	4
2 Inventory: what already exists	4
The record-keeping of the FFGFT corpus	4
The governance of Dot Theory	4
The Constitutional Interpretation Layer	4
F O R M and the Constitutional Verification Protocol	5
FER	5
Practice in audits	5
What has proved unsuitable	6
3 Proposal: ten building blocks	6
Graded depth of review	8
Origin of the building blocks	9
4 Limits and open points	9
5 Result	9

Peer Review under Present Conditions

Abstract. Classical peer review assumes that a referee can survey a paper in a few hours, that author and referee belong to an institution, and that texts are written by people. None of these assumptions holds in general any more: corpora run to hundreds of documents, independent researchers have no access to the usual submission routes, and language models produce plausible texts, scripts and apparent review reports in any quantity. This document collects the review tools that have already arisen in the exchange of the *Information Physics Institute* (IPI), in Stefaan Vossen's Dot Theory governance and in the FFGFT corpus's own record-keeping, and asks which of them can be used for a review framework suited to present conditions. The result is a proposal of ten building blocks: a declared constitution before review, an entry path provided by the author, testable objects instead of texts, predictions fixed in advance, separation of roles, a continuing register, rules for the use of language models and for simplifications, localised instead of global verdicts, and the explicit acknowledgement that there is no universal checker. The document contains no physical claims; it is a methodological inventory with the character of a proposal.

Legend

Label	Meaning
[K], [B], [S], [Q], [X]	FFGFT status markers: numerically confirmed, algebraically proved, structurally plausible with open derivation, taken from an external source, recognised as false
[SETZUNG]	explicitly posited foundational assumption, not derived
IPI	Information Physics Institute, mailing list and working groups
F O R M	<i>Framework – Operator – Residual – Model</i> , the grammar of the <i>Constitutional Verification Protocol</i> (CVP)
CIL	<i>Constitutional Interpretation Layer</i> (Vossen/Pascher)
CCR	<i>Canonical Constitutional Record of Constitutional Physics</i> (Vossen)
FER	analytical method of Diana Haskins (Universal Constraint)
OAP, FAH	admissibility record and <i>Framework Admissibility History</i> of Dot Theory
O0, O2	originator declaration and secondary rendering (Doc. 272)
SHA-256	cryptographic hash establishing the identity of a file object

1 Starting point

What the classical procedure assumes

Journal peer review is built for the article: a finished paper of ten to thirty pages, one referee from the same field, a few hours of reading, a verdict in three grades (accept, revise, reject). It rests on four tacit assumptions:

1. The work can be surveyed in one sitting.
2. The referee shares with the author a common stock of methods against which deviations stand out.
3. Institutional affiliation acts as a prior filter.
4. A text is the product of human work, so its length is a rough measure of the effort behind it.

What has changed

Scale. A framework such as FFGFT now comprises more than three hundred and seventy documents with their own status markers, a correction register (Doc. 190) and cross-references between geometry, Galois theory and particle physics. Nobody takes that in within a few hours. This is not a special case: every serious theory reaches this point, and established programmes do not open up in an afternoon either.

Access. Independent researchers have no institutional address; indexing services often list only what is already listed through publisher platforms. Repositories (GitHub) and archives with DOIs (Zenodo) are public and citable, but the review system hardly registers them.

Language models. Text, code and apparent review reports can now be produced at negligible cost. The length of a text no longer measures the work behind it. In the IPI exchange there have been cases in which a language-model prompt was submitted as "source code", in which a check block printed its own constants and then "PASS" beneath them, and in which a second language model with shared context reproduced an earlier analysis verbatim instead of checking it independently.

Pattern search. Any pattern can be read out of enough numbers. The image of the magic lamp that one rubs until something appears describes a real misuse of computing power. It does not describe every piece of work that contains numbers. The difference lies in the order: in pattern search the measured values stand at the beginning, in a derivation they stand at the end as the test. In FFGFT the geometry $T^4/\mathbb{Z}_3$ is fixed as a [SETZUNG]; from it follow algebraically $\mathbb{Z}_3$, $\text{GF}(27)$, D_4 and triality; only from these come numbers, and only afterwards are they compared with measurements (Doc. 365). An outsider sees at first glance only the number, not the order, and suspects the lamp.

Simplification and prejudice. A short entry path is possible and exists in several forms. It is, however, often read as faulty in itself, because the reader takes the simplification for the claim. Whoever comes expecting that work outside the institutions cannot be right finds the confirmation in every simplification. The burden therefore lies on both sides: on the author, who must build the entry path so that the order is visible, and on the reviewer, who must consult the full version before declaring an error.

The question

The question is not whether review is necessary but what it can look like under these conditions: who reviews what, on which object, in which order, with what binding force, and who retains interpretive authority over what.

2 Inventory: what already exists

The following tools were not designed for a review procedure. They arose in the work itself, often from a concrete dispute. That is precisely why they are tested.

The record-keeping of the FFGFT corpus

- **Status markers.** Every statement carries a status: [K], [B], [S], [Q], [X], [SETZUNG]. The reader sees at once what is proved, what is computed, what is open and what is taken over. The markers existed before the CCR correspondence; later terminology describes what they do (Doc. 351, methodological note).
- **Correction register.** Doc. 190 records corrections to earlier documents and the closing of points held open, continuously; earlier versions are frozen as an archive. Corrections are not worked in silently; they are recorded.
- **Check scripts.** Every recent document comes with a script containing assertions that prints PASS or FAIL for each statement. Many of these scripts are null tests: if chance yields the same result, it is not counted.
- **Negative results as sections.** Doc. 342 explicitly grants Galois products with $k \geq 5$ no status from product search; Doc. 361 has a section of its own, "What does not match"; Doc. 276 records a weak result as an honest negative.
- **Guard-rail against exponent numerology.** Every quantity can be written as ξ^N ; a measured scale inside an exponent is not a prediction (Doc. 190, P35).
- **Comparator is not input.** Measured values (PDG) serve for comparison, not for derivation. This separation has been taken into F O R M as Illustration 001.

The governance of Dot Theory

Dot Theory (Vossen) is by its own account not a competing physical model but a representational meta-programme. Its governance layer is set out in Doc. 272: *Lexicon* (terms), *OAP* (admissibility), *Matrix* (relations between frameworks), *FAH* (how admissibility changes). Two distinctions carry it:

- *Preservation of meaning-making versus allocation of meaning:* recording how a result was understood is admissible; deciding who is right is not.
- *Originator declaration (O0) versus secondary rendering (O2):* a third party's reading of a framework is marked as such; the right of correction stays with the originator. Records are permanent and contestable; later entries may refine, dispute or supersede, but the earlier rendering remains part of the history.

The Constitutional Interpretation Layer

The CIL is co-authored by Stefaan Vossen and the author; the co-authorship is limited to the CIL specification and does not extend to Dot Theory or the CCR. Its subject is not

the correctness of scientific statements but the entitlement to draw, inherit and pass on conclusions. Explicitly excluded are scientific correctness, empirical validity, mathematical truth and ontological commitment. From the dialogue between the two authors came the principles

- 3 – *Asymptotic Humility*,
- 6 – *Boundary Legitimacy / Evidential Admissibility*,
- 7 – *Recoverability / Reproducibility*,
- 8 – *Asserted Constitutional Properties*: an asserted property (determinism, self-management, derivational status) remains an authorial assertion until it is independently recoverable.

Further principles of version v1.1 name the vocabulary trap (adopting the vocabulary is no evidence of adopting the method; similarity of result or terminology is not a bridge) and *constitutional drift* (undeclared extension of scope). The author added the condition that scope must be declared at the point of derivation and cannot be asserted retrospectively in a dispute; this turns the principle into a checkable test.

F O R M and the Constitutional Verification Protocol

F O R M (ed. Vossen; authors Haskins, Leavitt, Pascher, Quílez Zamora, Vossen; v1.0, fixed by SHA-256 bd2a1f2d...e8e90c0e) divides every review into *Framework* (the declared distinction space), *Operator* (the operation carried out), *Residual* (what survives the operation) and *Model* (the recoverable representation). Filling the R position with *Residual* rather than *Reality* goes back to a proposal of the author: *Reality* would have introduced an ontological commitment that the protocol is meant to avoid.

The argument against a universal checker comes from the same correspondence. A proof checker such as Lean works because axioms, type system and admissible operations were agreed beforehand; the checker comes second. The law works the same way: an agreed, contestable procedure precedes the ruling. Programming languages lie in between and make the point most sharply: they are strongly normalised, and yet different libraries return different results for the same integral, and $0.1 + 0.2 = 0.3$ is false in IEEE 754 arithmetic and true in exact rational arithmetic. If the result depends on the frame even there, then for heterogeneous scientific frameworks a single checker cannot be the goal, only a procedure that discloses the normalisation in each case and makes it recoverable.

FER

Diana Haskins' analytical method FER asks of every term what earned it its status, and does not let terms become stronger because they fit together nicely. An application of FER in the IPI exchange of 24–25 September 2026 produced a recursive review sequence: declared frame → operation → distinctions obtained → representation → next operation → recovery test → retained, lost or newly exposed distinction → revised representation. Three cases are kept apart: a distinction was available and is no longer recoverable; it was never represented; it was not recognised, but the information was present and a later operation exposes it. The representation is thus not the end of a review but the next object of review.

Practice in audits

Several review procedures of recent months have produced rules that hold beyond their occasion:

- **Freezing and hashing.** The author freezes the object and states its SHA-256 hash; the reviewer confirms the hash independently on the object received before reviewing content. In the SIM2 procedure two files with the same title, the same date and the same version string could be told apart only by their hashes.
- **Final standing is assigned, not claimed.** A submission that certifies itself "ADMISSIBLE" has no standing thereby. Standing is assigned by the re-audit.
- **Order.** Declare → freeze → implement → execute → audit independently → assign standing. In the Block II procedure (MOTHER-GEA, 25–27 September 2026) implementations repeatedly arrived before the specification; that is exactly where the errors occurred.
- **Two constitutional invariants** from the same procedure: *solver reports convergence* ≠ *stationary point* (an absolute solver tolerance is not a stationarity test; what is needed are scale-normalised residuals evaluated independently after the solve) and *changed frozen input* ≠ *same object*. A third finding was added: the same frozen inputs with two different potential functions yield two different objects; the specification must fix the function itself, not only its parameters.
- **Separation of roles.** In the CTA-FUSION declaration the author declared his complete exposure to his own corpus and thereby excluded himself from independent review of its fidelity. In the SIM2 procedure (Krüger) custodian of the blind tasks, attestor and builder are separate roles with declarations of interest; co-authors are not regarded as neutral.
- **Sealed advance prediction.** The prediction that Krüger's OP8-Neuro operator U3 adds no structure beyond the Markov limit was deposited with a hash before the computation and confirmed by the result.
- **Localise rather than judge.** A limitation in framework A establishes or refutes nothing about framework B. An author may decline a request for deeper disclosure for legitimate reasons; the only consequence is that independent recoverability ends at the declared boundary. That localises the standing; it does not condemn.

What has proved unsuitable

- **Several language-model referees from the same model.** Different roles or prompts increase coverage, not independence: one model has one set of blind spots. A second model with shared context is an echo, not a replication.
- **Self-confirming check blocks.** A block that compares its own constants with themselves cannot fail and is therefore not a check.
- **A hash over the output instead of the source.** It shows reproducibility of the output, not identity of the program and not physical correctness.
- **Asserted properties.** "Self-managing", "parameter-free", "frozen" are properties that must be shown; a single fitted constant in the description is enough to refute "parameter-free".

3 Proposal: ten building blocks

The following building blocks are derived from the inventory. They are not a finished framework but a proposal; a framework of rules is itself a posit and must be agreed by those who apply it.

B1 – Declared constitution before any review

The author discloses what is posited and what is derived, in which order the layers follow one another, and which status each statement has. For FFGFT this means: geometry → algebra → number → independent empirical comparison. This order belongs before the first number in every presentation.

B2 – An entry path provided by the author

Because no outsider will spend the time on the whole corpus, the burden lies with the author: a designated entry path that shows within about an hour what everything rests on and where it could fail. For FFGFT these are Doc. 365 (the foundation in four layers), Doc. 377 (the interfaces to other frameworks) and the fixed predictions, for instance $M_W = 80.420 \text{ GeV}$ and $\sin^2 \theta_{23} \in \{4/9, 5/9\}$ (Doc. 372).

B3 – Testable objects instead of texts

What is reviewed is a frozen object: document, script, input data, each with a hash over the source. A check script must be able to fail; it compares against an external data set fixed beforehand, aborts on missing identity instead of printing a substitute value, and contains null tests. Where a function is constitutive, the function itself is fixed, not only its parameters.

B4 – Predictions fixed in advance

What a framework risks is deposited before the comparison with data, with date and hash. A framework that fits only afterwards has risked nothing. The size of a corpus is no substitute for a single sharp prediction.

B5 – Separation of roles and interpretive authority

The author retains interpretive authority over his posits and their declared meaning (O0). He has no authority over whether a declared operation is actually carried out and whether the result follows; that is decided by review. Author, reviewer and, where needed, custodian and attestor are separate roles; each role declares its exposure. A third party's reading of a framework is marked as a secondary rendering.

B6 – A continuing register

Reviewer findings, author responses and corrections are recorded continuously and permanently, not overwritten. A withdrawn statement remains visible as withdrawn. Standing is assigned, not claimed, and every assignment is contestable.

B7 – Language models as tools, not as verdicts

The use of language models is disclosed. Their results are not adopted unchecked but secured by assertions against computation and source. A language-model review does not replace an independent judgement; several roles of the same model are not a majority. Responsibility for every statement remains with the person who signs it.

B8 – Mark simplification as such

Simplified presentations are marked as such and name the full version. An objection to the simplification is not an objection to the derivation; before a reviewer declares an error, he consults the full version.

B9 – Localise rather than judge globally

The outcome of a review is a status for each statement (for instance confirmed, not accepted, candidate), not an overall verdict on a framework. A boundary of recoverability localises the standing of what lies beyond it; it does not condemn it. Similar vocabulary does not establish a bridge between frameworks.

B10 – No universal checker

Every review presupposes an agreed normalisation, and that is itself a posit. The aim is not one checker for all frameworks but a procedure that discloses the normalisation in each case, separates inherited conventions (computing library, system of units) from framework-specific assumptions, and makes both recoverable.

Graded depth of review

To answer the question of time in practice, review can be divided into three stages that can be carried out one after another and by different people:

Stage	Question	Object and effort
1 Entry	Are posits, order and status markers declared? Are there predictions fixed in advance?	entry documents; about an hour
2 Object	Do the hashes match? Do the scripts run? Can their checks fail? Do the inputs agree with the specification?	frozen scripts and data; hours
3 Depth	Does a particular chain actually follow from the posits?	a single derivation with its predecessors; days

Stages 1 and 2 require no expertise in the subject matter of the framework and can largely be distributed. Stage 3 does require it and is realistic only for individual chains. A verdict at one stage holds only for that stage.

Origin of the building blocks

Block	Origin	Usability
B1	FFGFT status markers; F O R M (Frame-work)	directly
B2	FFGFT entry documents	directly, effort on the author's side
B3	audits (hash, scripts, Block II)	directly
B4	sealed advance prediction (OP8-Neuro)	directly
B5	Dot Theory O0/O2; CTA-FUSION; SIM2	roles can be filled only with several participants
B6	Doc. 190; FAH	directly
B7	experience in the IPI exchange	directly
B8	experience with entry texts	requires discipline on both sides
B9	CIL (scope, vocabulary trap); frame-work $A \neq B$	directly
B10	CVP correspondence (normalisation before checker)	as a principle, not as a procedure

4 Limits and open points

- **Effort.** Declaring, freezing, hashing and keeping the register take time. The Block II procedure showed that the procedure reliably makes errors visible; it also showed that many rounds may be needed before a reviewable object exists at all. Procedure cannot replace content.
- **Neutral reviewers are scarce.** In a small circle almost everyone involved is a co-author or correspondent. Separation of roles is then more a disclosure of closeness than real independence.
- **Recognition.** A framework of rules agreed within the IPI holds at first only there. Whether outsiders regard an object reviewed under these rules differently from an unreviewed one is open.
- **Prejudice cannot be regulated.** The building blocks make prejudice visible (for instance when an error verdict is issued without reference to the full version), but do not remove it.
- **Interpretive authority versus review finding.** The separation in B5 is clear in principle, not always in the individual case: whether a statement belongs to the posit or is meant to follow from it must itself be declared. Here the condition applies that scope must be fixed at the point of derivation.

5 Result

1. Classical peer review presupposes surveyability, a shared common stock, an institutional prior filter and human text production; none of these assumptions holds in general today.
2. In the IPI exchange, in Dot Theory governance, in CIL, F O R M and FER, and in the FFGFT record-keeping, tools have arisen that closed these gaps in concrete cases.
3. From them ten building blocks and a graded depth of review can be derived. Most can be applied directly; separation of roles and recognition depend on how many participants there are.
4. The framework of rules is itself a posit. It gains validity only through agreement by those who apply it, and like every posit it must be declared openly.

Check script: `2/python/Dok378_Skripte/pruef_378_verweise.py` (checks the cited corpus documents, figures and hashes)

Register (Doc. 190): no entry (new document, no correction).

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